

[< Corn](#)**AGRICULTURE**

Spider Mites

Banks grass mite: *Oligonychus pratensis*

Pacific spider mite: *Tetranychus pacificus*

Strawberry spider mite: *Tetranychus turkestanii*

Twospotted spider mite: *Tetranychus urticae*

Updated: 08/2008



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Description of the Pest

Mite infestations on corn frequently include a mixture of spider mite species, including [twospotted spider mite](#), Banks grass mite, [Pacific spider mite](#), and [strawberry spider mite](#). Of these mite species, twospotted spider mite and Pacific spider mite are most common.

Adult mites are about 0.06 inch in length, have four pairs of legs, are greenish to pink or cream colored, and have various sized black spots on the body. Under warm conditions spider mites move rapidly within the colony area. Spider mites have four stages of development: (1) the spherical, somewhat translucent [egg](#); (2) a six-legged translucent larval stage; (3) an eight-legged [nymphal stage](#); and (4) the eight-legged adult stage. A resting or quiescent stage occurs at the end of the larval and nymphal stages. A generation may pass in as few as 5 to 7 days in midsummer, or in a month during cool periods.

Damage

All active stages of spider mites damage corn by removing juices from infested leaves, causing premature drying that results in loss of leaf tissue, stalk breakage, and kernel shrinking. Damaged leaves become somewhat yellowish and stippled on the upper surface and grayish due to webbing on the undersurface. Spider mites can be a serious problem on corn, particularly silage and sweet corn.

Management

Keep spider mite populations in check by reducing dust and weed hosts and encouraging mite predators. If monitoring indicates a need, treatment may be necessary on mid-size corn.

Biological Control

Spider mite populations may be held at very low levels by a number of predatory insects and mites, particularly early in the season. [Thrips](#) are effective early season predators, feeding primarily on spider mite eggs. Spider mites provide an important food source for predators such as [minute pirate bugs](#) and [bigeyed bugs](#). Minimizing early season insecticide applications, which may reduce populations

of beneficials, will reduce spider mite outbreaks. Naturally occurring predatory mites exert some level of control. In some areas, releases of predatory mites have been used to manage spider mites in field corn. If predatory mites are to be released, be sure to release the appropriate predatory mite species for the area and time of year. Also use the correct release rate and the correct timing. Definitive guidelines have not been developed, but make releases before significant spider mite outbreaks occur.

Cultural Control

Reduce spider mite problems by keeping fields, field margins, and irrigation ditches clean of weed hosts. Spider mite populations may increase more rapidly in areas where dust deposits are heavy on corn leaves. Thus, reducing dust may reduce the spider mite problem.

Organically Acceptable Methods

Biological controls and cultural controls are acceptable to use in organically grown crops.

Monitoring

Infestations usually begin on the lower portions of the plants and move upward as mite numbers increase. Evaluating spider mite infestations is most efficient if randomly selected, older, lower leaves are picked and inspected for stippling on the upper surface and webbing, mites, and feeding scars on the lower surface. Spider mite infestations that reach the ear leaf are most damaging.

Treatment Decisions

If small colonies of spider mites are found on the lower leaves of young plants throughout the field, control may be cost effective. Treat when corn is 2 to 4 feet tall; applications made after the plants exceed 4 feet in height usually result in poor control because good coverage is difficult to obtain. Just treating a couple of swaths around

the field can keep spider mites from spreading into the remainder of the field.

Pesticides and Natural Enemy Releases

Not all registered pesticides are listed. **The following are ranked by their IPM value, with the most effective and least harmful to natural enemies, honey bees, and the [environment](#) listed at the top of the table.** When choosing a pesticide, consider information related to water and [air quality](#), resistance management, and the pesticide's properties and application timing. Use [PestManage](#) to compare summarized management options for different pests in the same crop. Always carefully read the label of the product being used and take all necessary [precautions when handling pesticides](#).



 [Expand table](#)

Rank 	Active ingredient 	Example trade name 
Application Timing Varies (See <i>UC IPM Pest Management</i>		
A	spiromesifen	Oberon 2 SC
B	propargite	Comite

— No information.

 Do not apply or allow to drift to plants that are flowering including weeds. Do not allow pesticide to contaminate water accessible to bees including puddles.

Do not apply or allow to drift to plants that are flowering including weeds, except when the application is made



between sunset and midnight if allowed by the pesticide label and regulations. Do not allow pesticide to contaminate water accessible to bees including puddles.



No bee precaution, except when required by the pesticide label or regulations.

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- ‡ ^a ^b Restricted entry interval (REI) is the number of hours (unless otherwise noted) from treatment until the treated area can be safely entered without personal protective equipment. Preharvest interval (PHI) is the number of days from treatment to harvest. In some cases, the REI exceeds the PHI. The longer of the two intervals is the minimum time that must elapse before harvest.
 - ¹ [↩] Group numbers for insecticides and miticides are assigned by the [Insecticide Resistance Action Committee](#) (IRAC). Insecticides with unknown modes of action are assigned mode-of-action group numbers (MoAs) that begin with UN. Rotate pesticides with a different mode-of-action group number, and do not use products with the same mode-of-action group number more than twice per season to help prevent the development of resistance. For example, the organophosphates have a group number of 1B; insecticides with a 1B group number should be alternated with insecticides that have a group number other than 1B.
 - ² [↩] Range of insect and mite groups affected by a pesticide. **Broad** means the pesticide affects most groups of insects and mites; **narrow** means the pesticide affects only a few specific groups.
 - ³ [↩] Risk of harm to honey bees. For more information, see [Bee Precaution Pesticide Ratings](#).
 - ⁴ [↩] Risk of harm to predatory mites. Toxicities are generally to western predatory mite, *Galendromus occidentalis*. Where differences have been measured in toxicity of the pesticide-resistant strain versus the native strain, these are listed as pesticide-resistant strain or native strain.

- ⁵ [a b](#) Risk of harm to parasitoids and general predators. Toxicities are averages of reported effects and should be used only as a general guide. Actual toxicity of a specific insecticide depends on factors including the application rate, environmental conditions, and the life stage and species of a parasitoid or predator.
- ⁶ [↩](#) Length of time residue affects natural enemies. **Short** means hours to days; **moderate** means days to 2 weeks; and **long** means many weeks or months.
- ⁷ [↩](#) Risk of harm to fish from leaching, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- ⁸ [↩](#) Risk of harm to fish from adsorbed runoff, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- ⁹ [↩](#) Risk of harm to fish from solution runoff, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- ¹⁰ [↩](#) Risk of harm to humans from leaching, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- ¹¹ [↩](#) Risk of harm to humans from solution runoff, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- ¹² [↩](#) Date information was last updated in the *UC IPM Pest Management Guidelines*.
- ¹³ [↩](#) Use lowest rates for best management of predatory mite to spider mite ratio.



UC IPM Pest Management Guidelines: Corn

UC ANR Publication 3443

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